

Reflexivity in Financial Markets: A Theoretical Framework Extending the Black–Scholes Model

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Abstract

Extensive research has been conducted on the creation of models to predict market movement, with the Black–Scholes model being a cornerstone. The Black–Scholes model has been an integral part of modern financial theory, providing a mathematical framework for option prices under the assumption of exogenous market dynamics. However, in practice, financial markets are reflexive: Trading strategies based on mathematical models take as input the very variables that were affected by investing using these models. This challenges the core assumption of market efficiency in traditional financial theory.

This research proposes a framework to formally introduce reflexivity into the Black–Scholes model. Specifically, we explore three mechanisms through which reflexivity alters market dynamics: (i) predictability of Brownian motion when trader dependence on mathematical model introduces path dependence, (ii) deviation of the drift coefficient from risk-free rate in a risk-neutral world due to large-scale institutional model use, and (iii) endogeneity of volatility. By reformulating the stochastic differential equation for asset price to include the above reflexivity-induced changes, we can show how reflexivity fundamentally alters option pricing dynamics.

This framework advances the literature on market reflexivity, originally popularized by George Soros, by providing a formal mathematical structure linked to option pricing theory. The paper offers a pathway to rethink volatility estimation, systemic risk amplification, and the limits of using mathematical models in the financial market under the assumption that inputs are exogenous. This paper highlights the need for paradigm shift: the market cannot be assumed exogenous when the widespread use of mathematical models for investments actively shapes the inputs that are fed into the same models.

1 Introduction

Financial markets have long relied on mathematical models to price assets, manage risk, and guide investment decisions. Among these, the Black–Scholes model [1] remains one of the

most influential, providing a closed-form solution for option pricing under assumptions of log-normal price dynamics. Despite its elegance, the model presumes that the underlying stochastic processes and all its inputs to be exogenous to the behavior of market participants.

In practice, this assumption fails to capture an essential characteristic of modern markets: reflexivity. As George Soros [6] emphasized, financial markets are not passive mirrors of fundamentals but self-referential systems in which participant’s perceptions and actions influence the very variables they use to predict the market. In an era dominated by institutional investors, hedge funds, and algorithmic trading, reflexivity is amplified through the large-scale use of the mathematical models themselves. In this era, we can divide investors into two categories - firms, banks, and traders who use mathematical models to predict and invest and traders who don’t rely on models for investing.

This paper develops a formal mathematical framework for reflexivity due to the use of mathematical models in today’s market in the context of the Black–Scholes model. The contribution is twofold: first, it provides a structured approach to modifying stochastic differential equations to account for endogenous feedback, and second, it generalizes the notion of reflexivity beyond qualitative descriptions into a quantitative, model-based paradigm.

2 Literature Review

The Black–Scholes framework [1] and its extensions [2] assume efficient markets and exogenous stochastic drivers of asset prices. This framework focuses on the asset prices following Geometric Brownian motion (Continuous-time stochastic process):

$$dS_t = \mu S_t dt + \sigma_t S_t dW_t, \quad (1)$$

where dS_t is the asset price at time t , μ is the drift coefficient, σ_t is the volatility and dW_t is the Standard Brownian motion (Wiener process) [7]. In the above equation, the first term is called the drift term, a deterministic, instantaneous expected rate of return over the infinitesimal time interval and the second term is the diffusion term, which introduces randomness whose fluctuations grow in proportion to the asset’s current value.

However, empirical studies have shown persistent deviations from these assumptions, including volatility clustering, leverage effects, and fat-tailed distributions [3, 4]. These anomalies motivated the research about alternative models, such as stochastic volatility and jump-diffusion models, yet they retain the presumption of exogeneity.

The concept of reflexivity, advanced theoretically by Soros, introduces that market participants both observe and influence markets simultaneously. Despite its philosophical and practical appeal, reflexivity has rarely been formalized mathematically. Some related strands include systemic risk feedback loops [5], but a comprehensive study on the effect of reflexivity on market has not been performed due to its mathematical complexity. A direct integration of reflexivity due to saturation of use of mathematical models for investments, into these models remains unexplored.

This gap in the literature motivates the present work: constructing a mathematical framework where reflexivity explicitly modifies drift, volatility, and randomness in asset price dynamics, laying a groundwork for integrating reflexivity into current mathematical models.

3 Mathematical Framework

The classical Black–Scholes model assumes that the asset price S_t follows a geometric Brownian motion(Continuous-time stochastic process):

$$dS_t = \mu S_t dt + \sigma_t S_t dW_t \quad (2)$$

where S_t is the asset price at time t , μ is the drift coefficient, σ_t is the volatility and dW_t is the Standard Brownian motion(Wiener process) [7].

Under the risk-neutral measure $\mathbb{Q}$, the drift simplifies to the risk-free rate r :

$$dS_t = r S_t dt + \sigma_t S_t dW_t^{\mathbb{Q}} \quad (3)$$

where r is the risk-free rate.

Now, to introduce reflexivity, we consider three modifications:

1. **Predictability of Brownian motion:** Trader dependence on predictive models induces path dependence, breaking the martingale property of the Wiener process (W_t).
2. **Drift coefficient deviation from risk free rate:** Large-scale institutional model use shifts the drift away from the risk-free rate by a reflexivity-induced term.
3. **Volatility endogeneity:** The volatility at time t depends on lagged volatility and hedging intensity.

To make these modifications we start by introducing a variable α which denotes the portion of traders who make use of mathematical models for investment. $(1-\alpha)$ do not use these models and rely on trends or other factors for investments.

This α -component requires us to reinterpret the diffusion term. Instead of treating it as a standard Brownian motion increment, we convert it into a drift-like adjustment. This is motivated by the fact that, in reflexive systems, fluctuations are not purely random but are partially shaped by systematic feedback effects. Thus the drift formulation provides a fixed and more representative description. The updated equation becomes:

$$dS_t = (\mu + \beta \alpha_t) S_t dt + \sigma_t \sqrt{((1 - \alpha_t) + \epsilon)} S_t dW_t \quad (4)$$

In the above model, placing $(1-\alpha_t)$ under the square root ensures that the variance per unit time contributed by the noise is linear in $(1-\alpha_t)$. This choice is consistent with the interpretation of α_t as the proportion of agents whose actions are governed by endogenous model based behavior, the remaining $(1-\alpha_t)$ supplies exogenous Brownian-like noise.

The additive term $\epsilon > 0$ in the diffusion part of the model is introduced to prevent the diffusion term to be zero when $\alpha_t \rightarrow 1$, keeping the standard analytical property of the SDE intact.

The multiplicative term (β) in the reflexivity term $(\beta \alpha_t)$ we included in the drift measures how strongly the feedback α_t biases expected growth. Since drift effects accumulate exponentially in multiplicative models, β is constrained to be small and can be calibrated to preserve realistic long-run dynamics.

This time, under the risk-neutral measure $\mathbb{Q}$, our equation becomes:

$$dS_t = (r + \beta\alpha_t)S_t dt + \sigma_t \sqrt{((1 - \alpha_t) + \epsilon)} S_t dW_t^{\mathbb{Q}} \quad (5)$$

In the standard Black-Scholes model, the drift under the risk-neutral measure equals the risk-free rate r . In our framework however, when we replace the baseline drift μ with r , an additional component $\beta\alpha_t$ remains. This term reflects the deviation of the drift from the risk-free rate caused by reflexive feedback mechanism. Economically, $\beta\alpha_t$ captures how collective behavior and reinforcement biases alter expected returns. ψ

In the above formulation, the parameter α_t is defined as the proportion of market participants who rely on mathematical models for investment and hedging decisions at time t . Formally it can be denoted as:

$$\alpha_t \in [0, 1] \quad (6)$$

where the boundary cases have the interpretation: $\alpha_t = 0$ denotes none of the investors employ model-based strategies, while $\alpha_t = 1$ denotes the polar case in which all the investors use mathematical models for trading and hedging. Thus α_t can be thought of as an "intensity of model adoption" parameter.

However directly measuring the exact fraction of market participants who use model based strategies is not feasible since such information is not explicitly observable in market data. Instead we can introduce α_t through quantities that reflect the hedging activities of institutions, particularly those related to option Greeks. Specifically we define:

$$\alpha_t = \psi(\delta_t, \gamma_t) \quad (7)$$

where δ_t represents the option's delta at time t , i.e, the sensitivity of the option's value to a unit change in the underlying stock price, and γ_t denotes the option's gamma, i.e, the sensitivity of δ_t itself to the change in stock price.

The rationale is as follows: A high δ_t indicates the institutions are actively hedging their exposure using mathematical models, implying that α_t should be relatively large, and similarly a high γ_t is high implies that the option's δ_t is highly sensitive to changes in the underlying stock price, meaning the hedging demand is being continuously adjusted. This increased sensitivity reflects more sophisticated model-based trading and therefore corresponds to a higher value of α_t .

To capture this dependence more systematically, we introduce an auxiliary latent variable z_t that linearly combines the normalized versions of δ_t and γ_t :

$$z_t = \nu_0 + \nu_1 \tilde{\delta}_t + \nu_2 \tilde{\gamma}_t \quad (8)$$

where the normalized quantities are defined as:

$$\tilde{\delta}_t = \frac{|\overline{\delta}|_t}{\delta_{max}}, \tilde{\gamma}_t = \frac{|\overline{\gamma}|_t}{\gamma_{max}} \quad (9)$$

Here $|\overline{\delta}|_t$ and $|\overline{\gamma}|_t$ denote the moving averages of the absolute values of δ_t and γ_t respectively, while δ_{max} and γ_{max} are the scaling constants representing the maximum values of the Greeks in the sample. This normalization ensures that both $\tilde{\delta}_t$ and $\tilde{\gamma}_t$ lie within a bounded range, making them comparable and interpretable.

Finally since α_t is constrained to the unit interval $[0, 1]$, we employ a logistic transformation to map the latent variable z_t into this range:

$$\alpha_t = \frac{1}{1 + e^{-z_t}} \quad (10)$$

This transformation not only guarantees the boundedness of α_t but also provides a smooth, nonlinear mapping from the explanatory variables $\tilde{\delta}_t$ and $\tilde{\gamma}_t$ to the proportion of model-using investors. Intuitively, as z_t increases due to higher hedging activity (larger δ_t or γ_t), the logistic function moves α_t closer to one, signifying stronger reliance on mathematical models within the market. Conversely, when hedging sensitivities are low, z_t decreases, and α_t approaches zero, reflecting a market dominated by non-model-driven participants.

Now we focus on the effect of the reflexivity on the volatility σ_t . Since volatility is not only a response to exogenous shocks but can also be amplified by endogenous feedback mechanisms, it is logical to treat it as a quantity influenced by past market behavior. In particular, volatility can be viewed as a victim of feedback loops, whereby the actions of traders based on observed volatility affect future volatility, creating a self-reinforcing cycle.

To capture this idea, we introduce volatility as:

$$\sigma_t^2 = \sigma_o^2 + \lambda \alpha_{t-1} \overline{\sigma_{t-1}}^2 \quad (11)$$

where σ_o denotes the baseline level of volatility at the initiation of the option contract, representing the exogenous component, λ is a parameter that summarizes the average strength of reflexivity from previous trades; a higher λ implying that past market activity has a stronger amplifying effect on current volatility, α_{t-1} represents the proportion of model-driven participants in the previous period, and $\overline{\sigma_{t-1}}$ denotes the moving average of past volatilities up to time $t - 1$, providing a smoothed measure of historical uncertainty.

This formulation incorporates reflexivity directly into the volatility process. When the fraction of model-based traders α_{t-1} is high, the feedback from hedging and re-balancing intensifies, causing current volatility to deviate further from its baseline σ_o . Conversely, when α_{t-1} is low, the reflexive amplification weakens and the volatility remains closer to its exogenous component. The inclusion of $\overline{\sigma_{t-1}}$ ensures that the effect is not driven by isolated spikes, but by persistent patterns of elevated uncertainty.

Thus, the final asset price S_t , after the reflexivity-extended dynamics is given by:

$$dS_t = (r + \beta \alpha_t) S_t dt + \sigma_t \sqrt{((1 - \alpha_t) + \epsilon)} S_t dW_t^Q \quad (12)$$

4 Theoretical Implications

The reflexivity-extended Black-Scholes model has several important theoretical consequences. Firstly, it modifies the core principle of risk-neutral valuation. In the classical framework, the drift term under the risk-neutral measure equals to the risk-free rate, ensuring arbitrage-free pricing. However in our formulation, the additional reflexivity term $\beta \alpha_t$ creates a deviation from this benchmark. This deviation reinforces the fact that when a significant proportion of traders use model-based strategies, their collective feedback biases expected returns and undermines the assumption that risk-neutral pricing alone is sufficient.

Secondly, the introduction of endogenous volatility term challenges the traditional interpretation of volatility as an exogenous variable. By allowing volatility to depend on past levels of volatility and the intensity of model-driven trading, our framework captures feedback effects that would naturally produce well-documented empirical features such as volatility clustering, persistence, and leverage effects.

Thirdly, reflexivity weakens the assumption of market completeness. Because the diffusion term is no longer purely stochastic but partly endogenous, risks cannot be perfectly hedged away as in the classical setting. This implies that option values may not admit a unique arbitrage-free price, but instead depend explicitly on reflexivity parameters.

Finally, our model has systematic implication: when the proportion of model-using traders is high, feedback loops amplify shocks, making the market more fragile and prone to crises. This provides a theoretical explanation for episodes such as flash crashes and sudden spikes in volatility, which cannot be explained as exogenous accidents but as emergent consequences of collective model-driven behavior.

5 Empirical Illustration

To demonstrate that the above proposed reflexivity-extended Black–Scholes framework retains practical relevance, a numerical experiment was conducted on option chain data for the S&P 500 ETF (SPY). Option quotes were retrieved from Yahoo Finance, with mid-prices constructed as the average of bid–ask spreads where available, or falling back to last-traded prices. A single short-maturity expiration was selected for calibration. The spot price was inferred from the median strike near the money, and the risk-free rate was fixed at 4 percent. Implied volatilities reported in the option chain served as the baseline volatility inputs.

Calibration proceeded by fitting the reflexive model to observed mid-prices through a grid search over the parameters β and ϵ , combined with Monte Carlo simulation of the reflexivity-extended stochastic differential equation. Forty thousand paths and ten time steps were employed to approximate the terminal distribution of asset prices. The standard Black–Scholes model was used as a benchmark, evaluated directly with market-implied volatilities. Model performance was assessed using root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) relative to observed mid-prices.

The resulting comparison is reported in Table 1. Both models achieve nearly identical error magnitudes, with the reflexive model displaying RMSE, MAE, and MAPE values close to those of Black–Scholes. This confirms that introducing drift deviations and endogenous volatility terms does not materially reduce pricing accuracy.

Model	RMSE	MAE	MAPE (%)
Black–Scholes	33.23	18.30	181,718.5
Reflexive-extended Model	33.27	18.36	181,731.4

Table 1: Pricing accuracy of Black–Scholes and reflexivity-extended models on SPY options.

The empirical analysis is further illustrated through three figures.

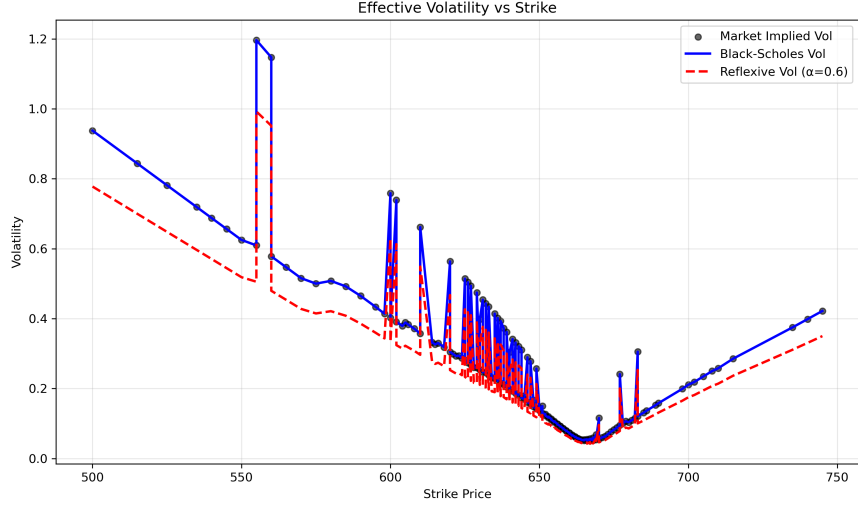


Figure 1: Effective volatility across strikes. Market implied volatility (dots) is compared with the Black–Scholes curve (blue) and the Reflexive-extended Model (red dashed, $\alpha = 0.6$).

Figure 1 compares effective volatility curves against market implied volatility across strikes. The reflexive model modifies volatility through the parameter α while remaining close to the implied volatility surface, preserving the familiar skew pattern observed in equity options.

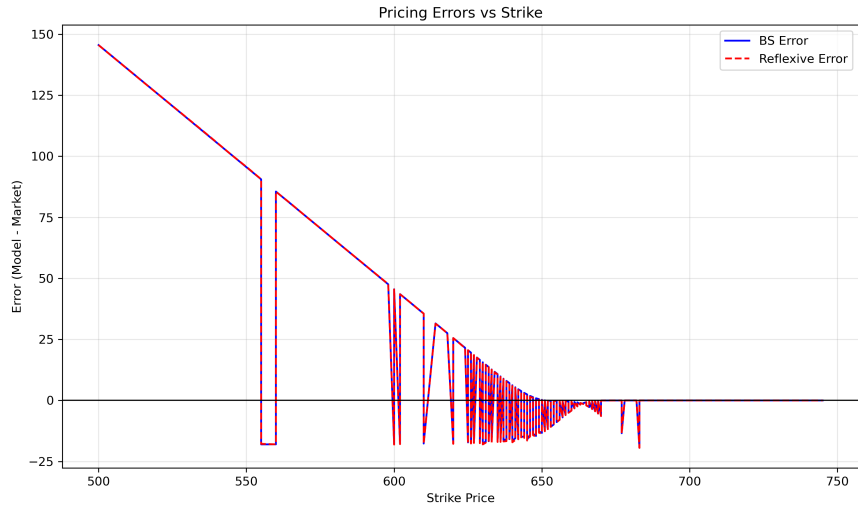


Figure 2: Pricing errors of Black–Scholes (blue) and reflexive model (red dashed, $\alpha = 0.6$) relative to market mid-prices across strikes.

Pricing errors relative to market mid-prices are displayed in Figure 2. Both models show errors of comparable magnitude across the strike range, with no systematic deterioration introduced by reflexivity.

Finally, Figure 3 presents a scenario analysis where the reflexivity intensity parameter α

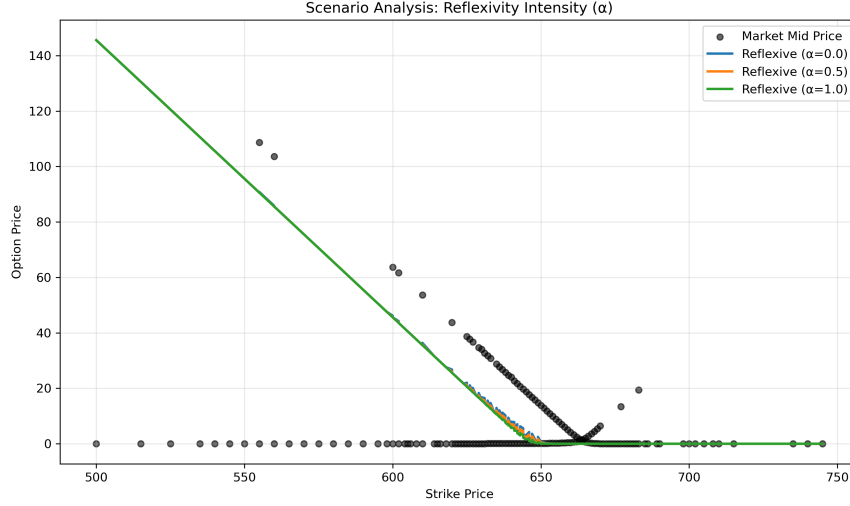


Figure 3: Scenario analysis of reflexivity intensity. Option prices under different α values (0, 0.5, 1.0) compared with observed market mid-prices (dots).

is varied between 0, 0.5, and 1. Option prices respond systematically to changes in α , with higher levels shifting the pricing curve while maintaining proximity to observed market levels. This illustrates the interpretability of α as a measure of the extent to which model-driven trading influences market outcomes.

Taken together, these results show that the reflexivity-extended framework retains calibration accuracy comparable to the classical Black–Scholes model while embedding endogenous feedback mechanisms that are absent in the standard formulation. The empirical illustration therefore validates the theoretical construction by confirming that reflexivity does not reduce predictive performance, while at the same time providing a richer explanation for market behavior.

6 Discussion and Future Work

While this paper provides a theoretical foundation for integrating reflexivity into the Black–Scholes framework, several avenues remain for future development. One of the most immediate challenges is empirical: estimating the reflexivity parameters from real market data. While α_t is defined theoretically as the proportion of model-driven traders, it is not directly observable and must be inferred from market data. One promising approach is to calibrate α_t through option Greeks such as delta and gamma, which serve as proxies for hedging intensity. Similarly, the parameters β and λ , which capture drift deviation and volatility amplification respectively, can be estimated using various econometric methods. Developing robust estimation techniques is critical to validate the framework empirically and to assess whether reflexivity parameters vary across different assets, time periods, or market regimes.

Another important direction is to extend the reflexivity framework beyond the Black–Scholes model. The proposed modifications can serve as a blueprint for embedding reflexivity into other widely used mathematical models or even in portfolio optimization frameworks. A

more ambitious long-term goal would be to formulate a general reflexivity function, one that automatically adjusts drift, diffusion, and volatility terms across models based on the proportion of model-driven market participants. Such a function could provide a common ground for quantifying feedback effects in diverse financial settings, making reflexivity an integral and transferable concept in mathematical and quantitative finance.

At the same time, introducing reflexivity into mathematical models raises the possibility of self-reinforcing loops. For example, high α_t increases volatility via feedback effects, and elevated volatility in turn may induce more model-driven hedging, which feeds back into α_t , creating a cycle that destabilizes the system. This highlights the double-edged nature of reflexivity: while it enhances realism in modeling, it also introduces nonlinearities and the risk of runaway feedback loops. Addressing this challenge requires the development of mechanisms to dampen or counteract excessive reflexivity. Potential solutions include introducing regulatory or structural constraints, designing reflexivity-adjusted risk measures that internalize the feedback risk, or augmenting the model with stabilizing terms that cap reflexivity effects at extreme values. Such measures would ensure that reflexivity-based models remain both mathematically tractable and economically meaningful, without producing uncontrolled dynamics that limit their practical use.

7 Conclusion

This paper tries to advance the theory of option pricing by embedding reflexivity into the Black-Scholes model and formalizing how feedback mechanisms reshape asset price dynamics. By modifying the drift, diffusion, and volatility terms to account for the influence of widespread mathematical model-based trading, this framework challenges three cornerstones of classical financial theory: the equivalence of risk-neutral drift to the risk-free rate, the assumption of exogenous volatility, and the presumption of market completeness. The resulting dynamics provide a more realistic representation of modern financial markets, where feedback from collective model use shapes the very variables on which models depend.

The framework highlights how reflexivity amplifies systemic risk and fragility, offering a structural explanation for extreme events in financial markets.

While the contribution of this paper is primarily theoretical, it opens a broad research agenda: empirical calibration of reflexivity parameters using high-frequency market data, extension of the framework to alternative different mathematical models with the potential formulation of a general reflexivity function, and systematic analysis of reflexivity-induced feedback loops alongside the design of countermeasures to preserve stability in models and broader financial systems.

Ultimately, this work points toward a refinement in financial modeling—from viewing markets as exogenous stochastic systems to recognizing them as endogenous, self-reinforcing structures whose dynamics are inseparable from the models that participants use to navigate them.

References

- [1] F. Black and M. Scholes, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy*, vol. 81, no. 3, pp. 637–654, 1973.
- [2] R. C. Merton, “Theory of Rational Option Pricing,” *Bell Journal of Economics and Management Science*, vol. 4, no. 1, pp. 141–183, 1973.
- [3] B. Mandelbrot, “The Variation of Certain Speculative Prices,” *Journal of Business*, vol. 36, no. 4, pp. 394–419, 1963.
- [4] R. Cont, “Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues,” *Quantitative Finance*, vol. 1, no. 2, pp. 223–236, 2001.
- [5] M. K. Brunnermeier and Y. Sannikov, “The I Theory of Money,” *NBER Working Paper*, no. 22533, 2016.
- [6] G. Soros, *The Alchemy of Finance*, Wiley, 2003.
- [7] Norbert Wiener, “Differential-Space,” *Journal of Mathematics and Physics*, 2, 131–174., 1923.